

Vacuous overflow

The six live by τ at finite N ; the branch grows; cofinal $\text{sch} < 0$ is open

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Abstract

Round 417 found $|\tau_{\text{un}}| \rightarrow 0$ (slope -0.47) with $|\text{sch}|$ flat, so $\text{sch} \rightarrow \varphi_{bb}$ and φ_{bb} would be the asymptotic sign carrier. Round 418 found six vacuous core windows with $\varphi_{bb} > 0$ that nevertheless live, via $\text{sch} = \varphi_{bb} - (\tilde{a}^2 + \tilde{b}^2)$. Those two cannot both be cofinal. Three hypotheses are tested. *H3* (the vacuous family empties) is **refuted**. A razor-pole account of the overflow $((C - I)^{-1}$ explosion at $C_{\min} - 1 \sim 10^{-7}$) is **refuted**. *H2* saves the six *now* ($\tau^2 > \varphi_{bb}$) but cannot be cofinal (vacuous τ_{un} is the one that vanishes). *H1* is a large- N census, not a law on the six. No RH claim.

Status. Lemma 1 is **reduced**, not proved: cofinal vacuous $\text{sch} < 0$ cannot use a τ -floor and cannot use branch-emptying; it reduces to $\varphi_{bb} < 0$ on large- N vacuous windows (EXT/selected census). Ausgang H3_REFUTED / RAZOR_POLE_REFUTED / TAU_SAVE_FINITE / COFINAL_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The tension

On the vacuous chart $J_2 = I_2$,

$$\text{sch} = \varphi_{bb} - \tau^2, \quad \tau^2 = \tilde{a}^2 + \tilde{b}^2, \quad \varphi_{bb} = c_J + \Sigma, \quad (1)$$

with $c_J = \text{den} - 2$ and $\Sigma = 2s^T(C + I)(C - I)^{-1}s$ for $C \succ I$. Living overflow: $\varphi_{bb} > 0$ and $\tau^2 > \varphi_{bb}$. Dead vacuous: $\varphi_{bb} > 0$ and $\tau^2 < \varphi_{bb}$.

Lemma 1 (vacuous overflow). *On the vacuous branch, $\text{sch} < 0$ cofinally — either via $\varphi_{bb} < 0$ cofinally (*H1/H3*) or via a source-pure lower bound on the couplings (*H2*: $\tau^2 > \varphi_{bb}$ wherever $\varphi_{bb} > 0$).*

Verdict: REDUZIERT. *H3* is false. *H2* is true at finite N for the named six and false as a cofinal floor. *H1* does not hold as a trend on the six; it holds as a census on EXT/selected vacuous windows. No k_0 .

2 Resolution of *H1*, *H2*, *H3*

Proposition 1 (three hypotheses). *On the named census:*

- *H1* (the six trend through zero). The six stay $\varphi_{bb} > 0$ on $N \in [151, 540]$; $d\varphi_{bb}/d\log N = -0.003$ (flat). $k_z = 49$ at $N = 540$ is still $+0.0197$. Healthy vacuous windows already at $N = 344$ are negative: the N -ranges overlap. Not a small- N artefact. EXT vacuous ($k_z = 70, 42, 72$) and selected vacuous ($k_z = 26, 43$) have $\varphi_{bb} < 0$ — a large- N census, not a law on the six.

- *H2 (vacuous τ does not vanish; r417 was P1-dominated). The opposite: vacuous $|\tau|$ slope -1.24 , $|\tau_{\text{un}}|$ slope -0.99 ; P1 $|\tau|$ slope $+0.05$, $|\tau_{\text{un}}|$ slope $+0.03$ (flat). The r417 fall is vacuous-driven. The six live now by $\tau^2 > \varphi_{bb}$ (margins 0.0015 – 0.046). A cofinal lower bound on τ fails because vacuous τ_{un} is the vanishing one.*
- *H3 (vacuous windows migrate to P1). Core vacuous $14/42$. Terciles by N : $3/14$, $4/14$, $7/14$ — the fraction grows. EXT $3/6$ vacuous. Selected mixed. **Refuted.***

3 What distinguishes the six

The overflow set is $k_z \in \{12, 13, 16, 19, 39, 49\}$. $k_z = 12$ is the r377 LATE prototype.

Proposition 2 (not a nearer pole). *Every vacuous core window sits at $C_{\min} - 1 \in [1.3 \cdot 10^{-7}, 7.5 \cdot 10^{-5}]$. Healthy windows are closer to the razor than overflow windows. The overlap of the Sherman–Morrison border s with the $C_{\min}$ -mode is $\sim 10^{-10}$; the razor share of Σ is at most 0.022 ($k_z = 39$), else ~ 0 . Overflow is not an explosion of $(C - I)^{-1}$. The r377 product $h_N h_{N+1} = +1$ on every vacuous window (overflow and healthy) — not a separator. $c_J \in [-0.520, -0.348]$ and $\Sigma \in [0.369, 0.650]$ on the six; $\varphi_{bb} = \Sigma + c_J$ is a small positive $O(0.01)$ (except $k_z = 39$, $\varphi_{bb} = +0.129$).*

The six are distinguished by $\Sigma > |c_J|$ together with a larger $|\tau_{\text{un}}|$ (0.068 – 0.399 against healthy 0.020 – 0.102), not by $C_{\min}$, not by r377, not by a razor pole.

4 Life versus death

Proposition 3 (the separator). *Living overflow: $\tau^2 > \varphi_{bb}$ (all six). Dead vacuous χ (χ_3 -23, χ_3 -33, χ_4 -20): $\varphi_{bb} > 0$ and $\tau^2 < \varphi_{bb}$, so $\text{sch} > 0$. That is the life/death mechanism of the overflow class. Dead vacuous $C_{\min} - 1$ is larger (10^{-4} – $3 \cdot 10^{-3}$) — further from the razor. Dropping τ on a living overflow sends sch to $\varphi_{bb} > 0$ (the mutant dies). Over $\mathbb{Q}$, $\varphi_{bb} = \frac{1}{10}$, $\tau^2 = \frac{1}{4}$ gives $\text{sch} = -\frac{3}{20}$; the drop- τ mutant is $+\frac{1}{10}$.*

5 Kills

Drop-border: no sch scalar ($\text{ind}_- K_2 = \text{ind}_- A_0$). Scramble $\text{nneg}(A_0) = 21$; two-period $S = 21$ has $\text{nneg} = 4$. The drop- τ mutant kills every living overflow. A false c_J is inherited from r418 (kills the φ_{bb} split, not the τ -save).

6 What is a theorem, what is a census

Equation (1) is a theorem (r417 chart). Proposition 3 is a finite identity plus a named separator. Lemma 1 is not proved: H3 is refuted, a cofinal τ -floor is unavailable, and the six themselves do not cross $\varphi_{bb} = 0$. Cofinal vacuous $\text{sch} < 0$ reduces to $\varphi_{bb} < 0$ on the large- N vacuous windows that the EXT/selected census exhibits — a named open, no k_0 . The six are a named finite class, not a cofinal phenomenon. No RH claim.